

Practical - 5

Aim:

Experiments on Packet capture tool
Wireshark.

Packet Sniffer:

* Sniffs messages being sent / received from
by your computer.

* Store and display the contents of the
various Protocol.

* Passive Program:

- never sends Packet itself
- no Packets addressed to it
- receives a copy of all Packet

Packet Sniffer Structure Diagnostic tools:

* TCP dump

- Eg. tcpdump -x host 10.129.141.24
xxx3.out

* Wireshark: - wireshark -r xxx3.out



Packet
Analyzer.

application

Application
www. Prom...

Copy of
all

Packet
capture
(PCAP)

Transport
network
link (Ethernet)
Physical

Description

Wireshark:

Wireshark a network analysis tool
formerly known as thread capture Packets
in real time and display them in human
readable format.

What we can do with Wireshark:

- * Capture network traffic
- * Decode packet Protocol using dissectors
- * Define filters - capture and display
- * Watch Smart Statistics
- * Analyse Problem

Capturing Packets:

After downloading and installing Wireshark
launch it and double check the name of a
network interface under capture to start
capturing Packets.

As seen as click the interface name you'll see
the packets start to appear in real time

The Packet list Pane.

The Packet list Pane displays all the Packets
in the current capture file. The 'Packet list'
Pane each line in the Packet list corresponds
to one Packet in the capture file.

The Packet Details Pane

The Packet details Pane shows the current
Packet in a more detailed form.

Capturing and analyzing Packets using Wireshark.

Procedure:

- * Select local Area Connection in Wireshark
- * Go to Capture → option
- * Select stop capture automatically after 100 Packets..
- * Then click Start capture
- * Save Packets.

1) Create a filter to display only TCP/UDP Packets, inspect the Packets and Provide flow graph.

Procedure:

- * Select local Area Connection in Wireshark
- * Go to capture → option
- * Select Stop capture automatically after 100 Packets
- * Then click Start capture
- * Search TCP Packets in Search bar.
- * To see flow Graph click Statistics → flow graph.
- * Save the Packets

2. Create a filter to display only ARP Packets and inspect the Packets.

- * Go to Capture → option
- * Select Stop capture automatically after 100 Packets

- * Then click Start capture
- * Search ARP Packets in Search bar.
- * Save the Packets.

3. Create a filter to display only one Packets and Provide the flow graph

- * Go to Capture → option
- * Select Stop capture automatically after 100 Packets.

* Then click Start capture

- * Search one Packets in Search bar.
- * To see flow Graph Statistics → Flow graph.
- * Save the Packets.

4) Create a filter to display only HTTP Packets and inspect the Packets.

- * Select Local Area Connection in Wireshark
- * Go to Capture → option
- * Select Stop capture automatically after 100 Packets
- * then click Start capture.

* Search HTTP Packets in Search bar.

* Save the Packets

5) Create a filter to display only IP/ICMP Packets and inspect Packets.

- * Select local Area Connection in Wireshark
- * Go to capture → option.

* Select Stop capture automatically after 100 Packets.

* Search ICMP/IP Packets in Search bar.

* Save the Packets.

Student observation:

1. What is Promiscuous mode?

A mode of operation of network interface card where it captures all packets on network, not just those addressed to it.

2. Does ARP Packet has transport layer header?

* No, ARP does not have a transport layer header.

* ARP works at network layer to map IP address to MAC address, so it does not use TCP/UDP.

3. Which transport layer Protocol is used by DNS?

* DNS mainly uses UDP (Port 53) for queries.

* It uses TCP (Port 53) for zone transfers or when responses are too large.

4. What is Port number used by HTTP Protocol?

The default Port number for HTTP is 80.

5. What is broadcast IP address?

* A special IP address used to send data to all nodes in a network.

Result:

Thus the experiment is complete successfully.

27/10/21